

Thesis Paper
Lee Meredith
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diamondsAndCode

diamondsAndCode

A Digital Dojo for DJs

by
Lee Meredith “tesuji”

Thesis Instructor: Marko Tandefelt
Thesis Writing Instructor: Loretta J. Wolozin

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Abstract

“diamondsAndCode” is pioneering new media by giving you control of more than just sound. It is also the first to use the turntable as a game controller and that game provides a quantifiable explanation of what the turntable does while giving an opportunity to improve your abilities.

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Chapter 1 **Introduction**

1.1 Concept

"diamondsAndCode" is a Open Source Augmented Reality performative teaching game for turntable practice and instrumentation, with the focus on the techniques of scratching, beat juggling and User-generated calibration/content. The game uses the exact position of the needle in the record as the quantifiable data that advances score and the win state.

"diamondsAndCode" biphasic system of a question then answer is traditional to music competition and maintained by DJ culture. This is audience independent, Procedural content generation iterated and does not require a pre-subscribed ability by player. It is familiar to the intended community and recognizable by beginners.

1.2 Audience and Setting

A great deal of intellectual property laws stifle a good amount of the cognitive load of the evolution of the turntable. It overshadowed the social contagion that had been based on New York's need to find an answer to local disputes and unrest. Live Art competition deserves a much longer thesis brief than I am prepared to write. But no other relationship was more fortuitous than the turntable and Contractarianism. Nowhere else this is as prevalent as within tradition battle style. Improving the system is at the heart of what could fix problems with communication and enthusiasm for performers.

1.3 Impetus

I had not been a patron to the scene very long until I realized that friends had held the key to some battles. This included bedrooms rooms to juried competition. If

there had been quiet moments in a performer set a or even worse conversation then things diminished quickly. But if there was a pointing and "OOOHHH" when you need it, the tide will turn and hide any other unoriginal style that would come after. Well I guess I felt a little cheated as a audience member. Here I see recognition fall into the wrong hands.

Much like the large percent of Americans I have lived where distance between me and everything had been significant. I do believe the Internet has improved the DJ learning environment. YouTube has lessons and some popular DJs charge money for online resources, So it is better than when I started. The Combining solutions in where having a visual representation augmented with the POV of the DJ related to a indisputable score that reflects within the social contract of the hip-hop narrative. It would need to be valuable enough from living rooms to music halls for learning and explanation of the art. This is the queries at the foundation that "diamondsAndCode" rest.

1.4 Design Questions

My most important question is how can I use computer visualizations to improve collaboration and understanding between musicians through using data from the turntable. The significance of this rests within the lack of a widely accepted musical notation. There has been intent to create a methodology as in [11] TTM turntable transcription methodology. The DJs normally chooses improvisation and recording equipment as an alternative to paper archiving. These issues are coupled with the preservation of its battle tradition in which DJs challenge each other by a call and response mechanic. When there are to few audience members to choose a winner by

cheering the loudest for the performer who they liked better, DJ's try to repeat the other musician previous scratch and when it is their turn to scratch they try to improve upon it beyond their opponents abilities. This is not only a way to decide who is better this is also an environment in which teaching and learning occur. "mentoring" By using game can I preserve and collect this appreciative form.

Most of my sub questions revolve around the implications of how to use open source products and does this facilitate programming to those who would normally not



figure 1: Ms. Pinky



figure 2: Dr. Gottfried

find themselves
doing so.
My subset of
questions have to do
with the transition
into augmented
reality and the
convergence of

performance / game. Will augmented reality enhance the user's experience. What things do I have to lose by being constrained by the environment in where you need a camera. Is there any mixture between camera independent images and those that are solely virtual.

The technical issues at this point separate twice. One is on the primary and the other being pulmonary. Primary is the [7] Ms. Pinky Interdimensional Wrecked' System (IWS) which was enabled by the invented by [4] Dr. Gottfried Ungerboeck at the IBM Zurich Research Laboratory in Switzerland. His invention is also known as "Trellis

Coded Modulation" that is the most optimal way to encode ones and zeros in on our walk waveform this allows for the maximum amount of data to be transferred over an analog telephone line. This minimized small noise and distortion. When pressed on vinyl records and transferred to the computer the zeros and ones can represent any media and can be manipulated as if the media is pressed on the vinyl itself. This is the way I am able to control the media in "diamondsAndCode".

The pulmonary technology at the beginning this had been [10] Processing and [9] OpenFrameworks. [1] ARToolKit plus is also as I earlier mentioned a possible method to nearly contextualize the turntable. By placing Homography recognizable tags on or next to the vinyl records and audio mixer I can give the appearance of virtual objects of being in the three-dimensional world through the ARToolKit Plus open source C++ library. The augmented reality toolkit was first placed on the Open Framework's form by Theo Watson and then modified by Zach Gage to include the option of more Homography recognizable tags.

Chapter 2 **Domains and Precedents**

2.1 DJ Battle

“diamondsAndCode” live in the intersection of rhythm games and turntable traditions of battle. The first precedence that I am mentioning is when two people who are DJ's want to discover who is better they proceed to get an response from the audience and have that audience vote by cheering loudest for the DJ for who they felt was the better. This comes from earlier blues tradition of cutting heads in wear the same is done but with guitar players. This relates because of the audiences desire to want to know who is better and how competitive play can also be a space for learning.

It is so widely expected and circumstances where there are no audiences sufficient enough to make the decision to musicians will try to copy each other one for one. Not only will they do what their opponent does they may add a phrase at the end of the one they just repeat it. This is to say their are good enough to move on beyond the challenge they received from their opponent. My own work relates in question of who was better and how my product functions. I am differentiating myself by finding the quantifiable data set to find out if I'm correct when their turn had been tested.

2.2 Audio-Surf

Other places where the same interaction had occurred in rhythm games. There is so many that I feel relates but the one that separates itself and relates the most is [2] Audio-Surf. What's sets it apart from other games is that it uses music that you supply as what it creates the rhythm roller coaster you are to collect points with in. “dAC” not

only says that participants are to create the challenge but they set bar to bar of music, so no two games will ever be the same. There are many games that help teach music theory but there are few that teach by using the actual instrument.

2.3 Real Instrument Games

An example of real instrument games is [5] Guitar Rising. It is limited to only playing single notes. It emulates traditional rhythm games and where there is a highway of notes that players must hit.



figure 5: Guitar Rising

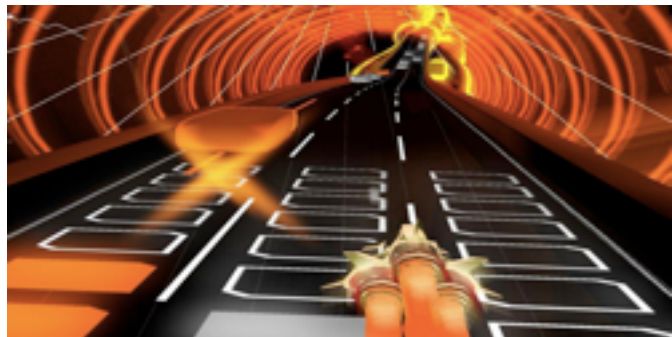


figure 4: Audio-Surf

Chapter 3 **Methodology**

3.1 Pong

My methodology revolves around iteration to specification and delineation. My first iterative experiment for “diamondsAndCode” looked at the difference between programming with similar open source resources and their capacity to communicate

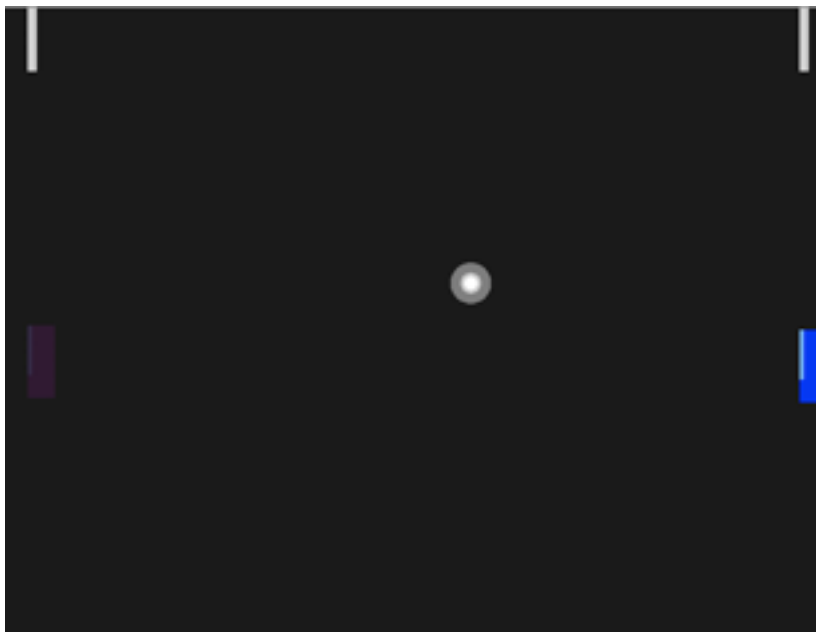


figure 3: musical version of "Pong"

over great distances by its affordability. I wanted to introduce collaborative gameplay with a individual score for each participant. I chose to make a technical exploratory path by taking code from Processing I had written into

OpenFrameworks a open

source project. I chose to make a musical version of "Pong" that had been popularized in the 70s by Allan Alcorn. In the version I created I included two paddles that step along the left and right, the same distance from the wall as the traditional paddles from the original game. The two sequence paddles represent a potential beat/song.

To collect points the players must be in the same coordinates as the moving paddle. The sound confirms the point has been collected but only by keeping the ball in play without letting it pass your paddle will the points you have collected be added to

your overall score. If you fail to keep the ball in play your potential points would be set to zero so even though you may have your half of the beat without continuing the following you don't succeed. Also it is my assumption that this will create a unique beat by the player being forced to leave randomly to add their potential to their overall score.

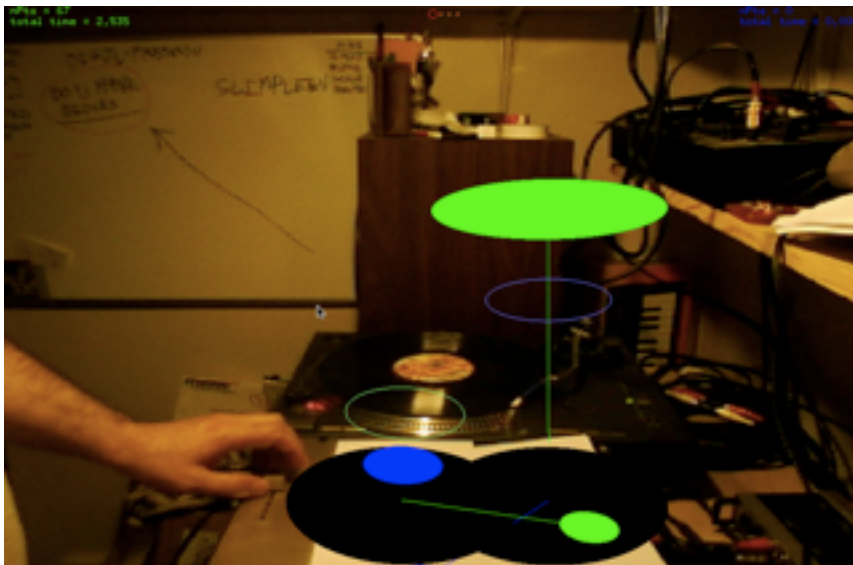


figure 9: Diamonds & Code as of Dec.21 2009

Through this focus creation I explore my inquiries in gameplay that emulates other artistic social contracts as in the battle for technical turntable supremacy is the same social contracts that games inhabit. By having even a broken

game that allows for the turntable as an interface can be user tested. This is paramount to what I consider to be necessary to begin a quantifiable process in which to make a good user experience. As I shared this first step I discovered that most said they wanted more explicit DJ narrative. The decision to continue with OpenFrameworks became a technical staple from here on out.

It is widely understood that the turntable, when used as an instrument can create unique, transformative music. Much like all instruments it is very simple interactions in which one embodies to create emotion and compelling sound. When these movements are expressed in series the phrasing becomes very complicated. "diamondsAndCode"

works to embody those simple components. I created a illustration of the goals for the final project and what I perceive the three states of the games mechanic.

3.2 First Person

This is an example uses a first person perspective on how the interface with the approach by two DJ's with disparaging differences between abilities and how the interface could accommodate them both as opponents for each other. This is a prescriptive functional explanation to myself and others, of the properties in which the turntable is used most frequently by musicians. I have included this documentation as Appendix First Person. The final project changes quite a bit from this illustrated explanation. This had been useful as a record of where I move a different direction.

3.4 User Scenario

The next part of the process I began isolating these components and exclude all of the augmented reality and instead constrain the next iteration the of the game to 2D graphics. Leaving out ARToolKit Plus had left me with the least amount of noise, so questions of costing and balancing the parts of the game for players were focused. In figure[] the circle represents where each player is during all recording or playing. I eliminated the idea of visually representing which person will be recording and playing. In the augmented reality version I go in and "Y" axis. In this iteration I move them over the top of each other perpendicular but place the tracing circle above the pre-recorded scratch the opponent is to test themselves against. This became the first opportunity to begin recording the sound of



figure 10: Cross fader

the scratching in tandem with points that was recorded at the same time and then played back.

After I created the 2D program I chose to port the code to be represented in three dimensions. As in Appendices UserScenario_0.0. what was being represented in my paper prototype had become actualized. At this point the system of depositing squares for direction and time spent being still has not been implemented in my final semester prototype and there is no programming in which the system checks correct answers. What is also missing from this prototype is the cross fader and checking if it is on or off while the scratching.



figure 6: first game test

3.5 Surveys

In my first game test I used 2 surveys as the foundation protocol in which I derive the answer in how they classify themselves as a player and or DJ. As well as their qualitative experience. I tested for their opinion of the graphics and it's role in the games mechanic of tracing the other players scratch. To this end I could have been better for the people who video games are not played. The first survey was redundant. This was due to too many similar questions leading to similar answers. As for example, if I asked about video games and they had already said they do not play video games, I was not getting any thing new.

I first gave page 1 of the survey and at that time I asked people to create a DJ name. This served two purposes the first is to be able to identify players on the survey and the second is the exploration of avatar creation for later iterations of the game. I also discovered that this is a good barometer for general interest. It appeared as much time was taken with the DJ name so went their enthusiasm after playing. I begin testing with the first people immediately after finishing page 1 of survey so not to waste time and make future participants the audience. I had hope to witness if there is any participation between audience and player as well as if there is any improvement by players who have watched before directly participating with the game. It did not elicit any behavior worth noting at this time. I felt this was probably due to the volume of users being low.

In the survey that they receive after they've tested I asked if they were able to trace the other players scratch with or without the visual representation and did they have a preference. The second survey also considers their opinions and general musical experience. The users wanted more visual clarity when it's their turn to record. A great deal of the information from the second survey is useful, although I can lead the questions better than I have.

Prior to doing the test I expected that this protocol could help find multilateral information on the subjects game experience. The second half of the survey gave voice to practical user opinion so to shape my focus. But the survey could be improved for the next time.

3.5 Scratch Academy

My second user test I had contacted [3] DJ scratch Academy which was created by Jason William Mizell more widely known as Jam-Master Jay the DJ of the influential hip-hop group Run-D.M.C. who had died in a mysterious murder in 2002.



figure 7: Jam-Master Jay

Though it took some persistence I was given the opportunity to show on the last day of their classes. I had to lay aside my expectations even if I had an improved survey. I had visited two other times regarding my opportunity for feedback. This has been new for them, that is to have outside person demonstrating their computerize DJ learning environment. Even though the whole point was to get the opinions of great DJ teachers it was frankly easier to know how to approach design and technology students with surveys.

I was prepared to do any one of the myriad things. I had canned footage of the color tracking as shown in figure 8: and a movie of the last example of the game I had made. Naturally I brought the entire set up. I was willing to do my keynote but thought that would be fairly stuffy.

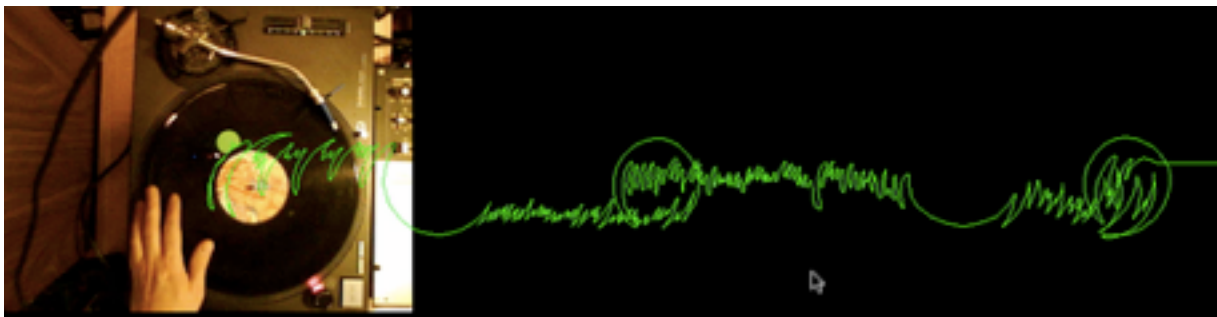


figure 8: Color Tracking

This is not mentioning the epic weather which coded the entire city with snow which caused a dearth of students to show for the three classes I attended. The weather also caused my contact to run an hour late leaving me no sense of direction for my own performance. But of course in tandem I had come to watch how the largest DJ school in the world pop their beginning classes to their advanced.

In the first hour I watched 101 class on beat matching and they were giving their final performances. After all of them have been up in front of each other with their performances that appear to last around seven minutes the head teacher and the two TAs gave criticism. At that time a student was asked to fulfill the homework they were given which had been to think of a DJ name. This had been a protocol I used in my first user test at the top of my surveys. This suggests to me a that this is ceremonious even though it was taken lightly. I will continue exploring ways to incorporate that ceremony in my own game.

At the end of the first class I attended I began showing my movies. The TA's from the first-class and another teacher from across which have been held at the same time in a small room at the back of the school. This gave me an understanding of the informality they were used to and spurred multiple conversations amongst themselves in what the users may possibly be. The color tracking at that time led to supporting the idea of TTM Turntable Transcription Methodology with this technology. These particular conversations also revolved around the issues of getting information from the cross fader. I took that as what professional DJs had wanted and how they saw capturing the cross fader information as needed for the future of any project that has interest in teaching or transcribing what DJs do. The second class I had watched had been more

scratching centric in where they reviewed scratches from previous classes and added new ones for them to practice in what they called their "free-for-all" day. They pumped room with one beat and everyone at their stations had individual speakers and where they could hear themselves practice. It had always been on my mind since I had first heard of the school how did they deal with scratching all at the same time. It seems simple and fairly useful.

At the end of that class I began setting up. All but two of the students left and the rest of the crowd consisted of the manager two teachers three TA's and the secretary. Those particular people knew what they were looking at more than anyone. They were very excited and said as much. Though it had not been directly discussed that my goals had been to create a learning environment it had been said that it would be helpful in teaching. I felt this was a great achievement but I had technical difficulties which no one noticed on the half of it being such a new way of seeing scratching. So I have been pretty discouraged on the technical side. The technical difficulties left me only showing the work and not allowing people the opportunity to use it as much as I had hoped. For some reason left turntable did not work. I had problems with sending audio. I have been invited back and plan to. For my next playtest I want feedback to come from the software on how well you are doing. Equally important is to have a mid-level user group. I would say that that would give me better answers on usability. That is also the group who I've yet to have any interactions with.

3.5 Game Book

For the end of this chapter and as the final explanation of my thesis project I include the Appendix Game Book. From my testing in general it was a struggled with

comprising the to images into one. In figure # I had no historical or timeline but have the point of view of what DJ normally looked at. The figure # below has lots of history but doesn't do much telegraphing so it can be copy by the opponent. I believe I have found both worlds in the "dAC" game. I have been able to keep the point of view And the top screen handles the history.

Chapter 4 **Evaluation**

4.1 Success

A part of the project was automatically a success before I had done much to construct its present day evolution. This had been based on the lack of partnership and acceptance on the part of both communities which were necessary for the construction of "dAC". On behalf of a digital / Game design community it is fairly understood there is a media war a foot. But getting themselves to recognize that they are making weapons for this media war is theoretical at best. At worst it is defined as to be lowbrow. This is probably based on its own struggle to be accepted as an art form that rivals or at least stands head and shoulders to any other art form.

Convincing a DJ that they are warriors in a street level media battle is so easy as to be regrettably cliché and silently understood. Now ask those same DJs about the intersection of technology and turntables there is a good amount of skepticism. Equal to any other insecurity that these two mediums have in common the most prevalent are they are both considered a lesser form than their predecessors. Game design for not being art but being more popular in one application than anyone painting. It is so similar in the music world to say that DJs are not musicians even though they have changed the face of the way we see the world for every single living human on the planet. If only to say the loss of audience members for the music which came before hip-hop.

4.2 Significant

It is a rare turn for an instrument to move from analog to digital successfully. So only to see the turntable in that light is fairly remarkable especially taking in account its lifespan is relatively short and its significance to modern culture.

The single piece cast iron frame is the reason for the modern piano and came out of the Industrial Revolution. Equal to this is that the piano keyboard and the turntable maintain its interface through its digital transformation. Just as the keyboard interface offered new in him terpretations of what instrument could be as it had in the 80s so has the turntable been give that digital resurrection. Though there is some obvious reasons to continue collecting vinyl. The audience cannot tell the difference between virtual vinyl and its predecessor.

"diamondsAndCode" explores the opportunity to which few musical instruments have been granted as in automated transcription system and the personification and testing of the musician's ability instantly. This is the holy grail for the music information retrieval (MIR) community. This is based on the relative ease in which dynamic graphical models (DGMs) are possible and give stamina through its immutability and precision.

DiamondsAndCode will be of benefit to DJs to pursue systematic investigations of musical information. This includes the asking of factual questions about DJs and the testing of hypotheses about DJ musical organization.

4.3 The Benefit

How DJs benefit from using computers and "dAC" is by utilizing processing speed in which you are allowed productivity in surmising the differences and similarities between performers. The computer lets you assess problems that are otherwise impossible to correct. It is clear what sets apart "diamondsAndCode" is not that it has invented battling/live art competition. It is that the drivers seat has been given to so few instruments.

There is not one musician who would say that a mentor at some level had improved their abilities. Those same musicians would agree that some of the most difficult parts of teaching music as well as appreciation being improved by audiences is for the form having visual explanations that pertain to what they actually do.

4.4 The Future

"dAC" takes by far the largest leap to explore in the ecology based narrative of turntable interface music creation as in a intended way to prove your abilities as a DJ. I would rate myself as a novice if only for the romantic notion of learning the one thing continuously. Or if not to shy away from the responsibility of learning. Clearly on the recognition of what it would take as well as what I would ask interested participants going forward is to learn to program. From just a little bit of knowledge on the surface of the backend will allow individual strength. I have been playing on turntables for most of my adult life and expert or not I feel confident in having a great deal of knowledge as a fan, user, and a technician to support my inquiries.

I was able to support my design process mainly through the diagrams which have forced me to explain in a visual manner. Most of all my concrete decisions came through play testing with multiple level interests in a kaleidoscope of components but without the failed illustrations I don't believe I could peek and rinse my process for others.

I had not thrown away the importance of showing this piece three days in the gallery. It required me to stay with the project from opening to close. It was a good thing no matter what. I discovered opinions varied from component component. I had many experiences throughout the testing and where one person liked what another had

not. This had influenced my process to include preference as being more valuable than to solidify a rigid body experience. This is not to say there is no through line which I think is understood as you see in the appendices but it is worth acknowledging what had been working for one participant did not work for another. It will influence directions of the product. That may mean automation of these options as well as simple selection.

I made the choice to direct the future of this project into a few directions that have yet to be program. It is desirable to make a scene that would allow musicians to record and visualize XML files. This would give an enormous amount of power to teachers if they only had to send a small file to a student that would replace opponent in the current state of the game. The option also would eventually be a way to archive notation to share. I want to complete things that had been the loose ends as in a interfaith that controls navigation by use of turntables. I'm also looking at C language libraries that deciphers beats per minute of whatever samples are being imported by users.

It would be notable if this project could find a home in the DJ schools and I will continue pursuing that end.

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- [1] ARToolKit Plus http://studierstube.icg.tu-graz.ac.at/handheld_ar/artoolkitplus.php
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- [3] DJ scratch Academy <http://www.scratch.com/flash/>
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First Person

This is an example uses a first person perspective on how the interface with the approach by two DJ's with disparaging differences between abilities and how the interface could accommodate them both as opponents for each other. This is a prescriptive functional explanation to myself and others, of the properties in which the turntable is used most frequently by musicians.

Turntable one "T1" is selected by the "Novice user." I stand in front of the "diamondsAndCode(dAC)" setup as is shown in the visual example "Set up." At the bottom of the screen there is a calibration that allows me to select the levels of "Difficulty" I would need to achieve to receive positive feedback for correct answers I would give. This placement is shown to me in the "Calibration interface" example. I am able to surmise from the title next to the individual calibration GUI 2d sliders that there are three choices and the ability to shut them off with square buttons to the left of the number boxes. I feel that it is too difficult for me to use a percentage that I think maybe to great for my ability. I move the slider around and see that the number go's from 10% being the weakest to 100% being the most precise answers needed. I choose 30% in all categories, this does include "Cross fader," "Freeze/Rev" and "Movement." These mechanics of the game and their corresponding icons are shown in the "Play State" visual document. Movement over time is represented by a Green line except when you're going in reverse. Where I am in the present is shown to me by a semi transparent green circle that gets bigger or smaller as it moves relative to the velocities within the

duration of the recording of the turntable made before the "Play State." This circle moves in 3-D space as it had be recorded on Turntable two "T2" so to be able to Follow/ trace for what my overall percentage of correct answers are. It is being clarified in brackets with the corresponding name "Rec / Leader" as it appears in the "Play State" document. The center turntable diagram also on the "Play State" page shows that you're able to get a percentage based on your current placement in which the turntable had stopped or had reverse its direction during the recording of the opponent in which I am testing myself against. When I am to trace my answer against the movement created over time as well as where I was to stop and reverse direction also I am expected to turn the audio off and on. This is called "Cross fader" in the difficulty interface window. I was able to turn off this option or calibrate it for my ability as I had "Movement" and "Freeze/Reverse." To the right of the "Difficulty" pop-up window there are two more options "Play" and "Chances." One in the number box next to "Play" is how I was able to set myself to record first and my opponent chooses 2 in their corresponding number box. Making her responsible for being the one that must answer my recording by copying it is closely as they can. "Chances" refers to how many times it will repeat before the last time in which you are to be tested. As I respond in my turn I get audio reinforcement that I am answering correctly. I can set those sounds and options as I did in the "Calibration Interface" page. This interface window is left of the "Output" and "total" window. I am able to set the Volume and whether it is on. If I change the beat settings there will be change in music that I scratch or beat juggle over. I could leave the audio alone for the beat setting and only change the other two being

"Cross fader" and "Freeze/Reverse." If I feel that the "Cross fader" audio, that is played back for me to match against is too loud to hear the audio that is being achieved by tracing correctly opponent recorded "Freeze/Reverse" hit zones, I am able to turn it down. This logically delineates to any of the interdependent audio options. Now I have selected in tandem with my opponent the necessary options. Since I am the one to record first I am the player to start the game by pressing the space bar there is a countdown for one measure of music after that time has expired I am allowed record a measure of manipulation as it is shown in "Record State" documentation. A red line appears in the center of my half of the game screen as well as at this time there are "Freeze/Reverse" shapes are being deposited. After the game ends I'm able to see the percentage of answers and the total of them all. I conclude this user scenario. There are several ways to calibrate the system so to strengthen weaknesses, if only to measure your ability. This user scenario is to layout specific points of destination for my project.

Calbration Interface



On / Off

Percentage needed
to be correct

How many chances per turn

Players turn

Setting of one records first
Setting of two records second
If both are set to one the recording will happen simultaneously

☐ T1 Difficulty

☐ 30% ☐ Crossfader

☐ 30% ☐ Freeze/Rev

☐ 30% ☐ Movmnt

Play 1

Chances 1

☐ T2 Difficulty

☐ 90% ☐ Crossfader

☐ 95% ☐ Freeze/Rev

☐ 95% ☐ Movmnt

Play 1

Chances 2

☐ T1 Volume

☐ 127 ☐ Beats

☐ 50 ☐ Crossfader

☐ 50 ☐ Freeze/Rev

Out Put

☐ 93%

☐ 95%

☐ 70%

Total 84%

☐ T2 Volume

☐ 127 ☐ Beats

☐ 50 ☐ Crossfader

☐ 50 ☐ Freeze/Rev

Out Put

☐ 70%

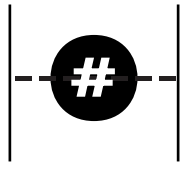
☐ 45%

☐ 70%

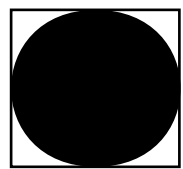
Total 65%

Rec State

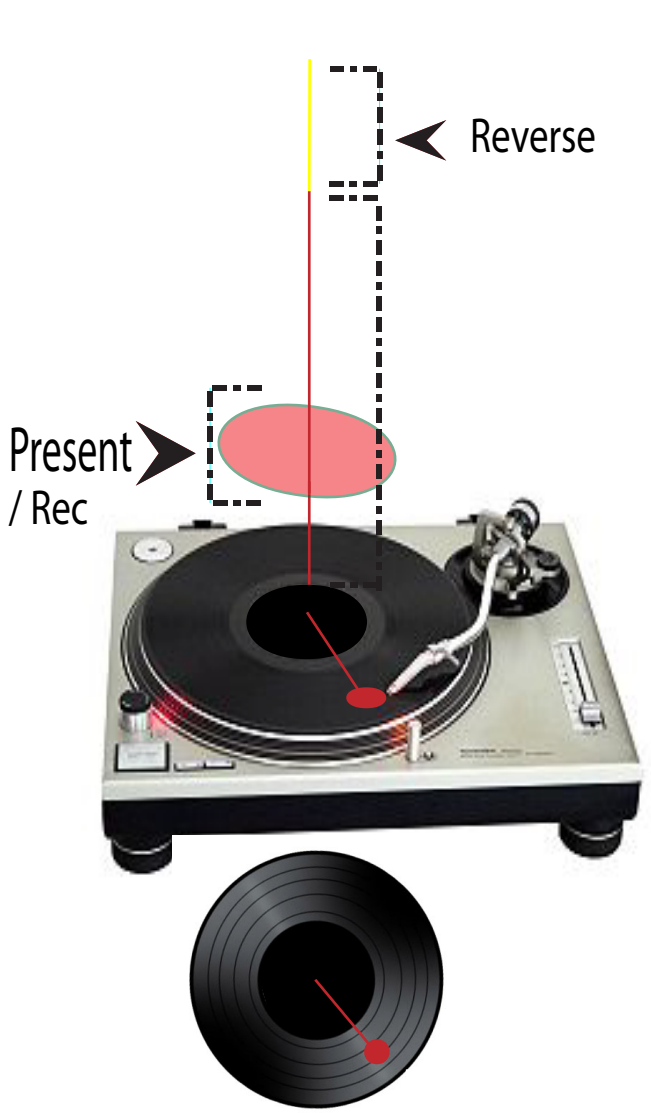
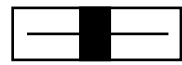
Movmnt Over Time



Freeze Or Reverse

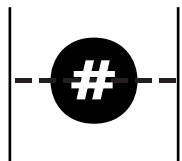


Audio Crossfader
On / Off

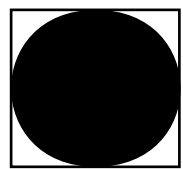


Play State

Movmnt Over Time

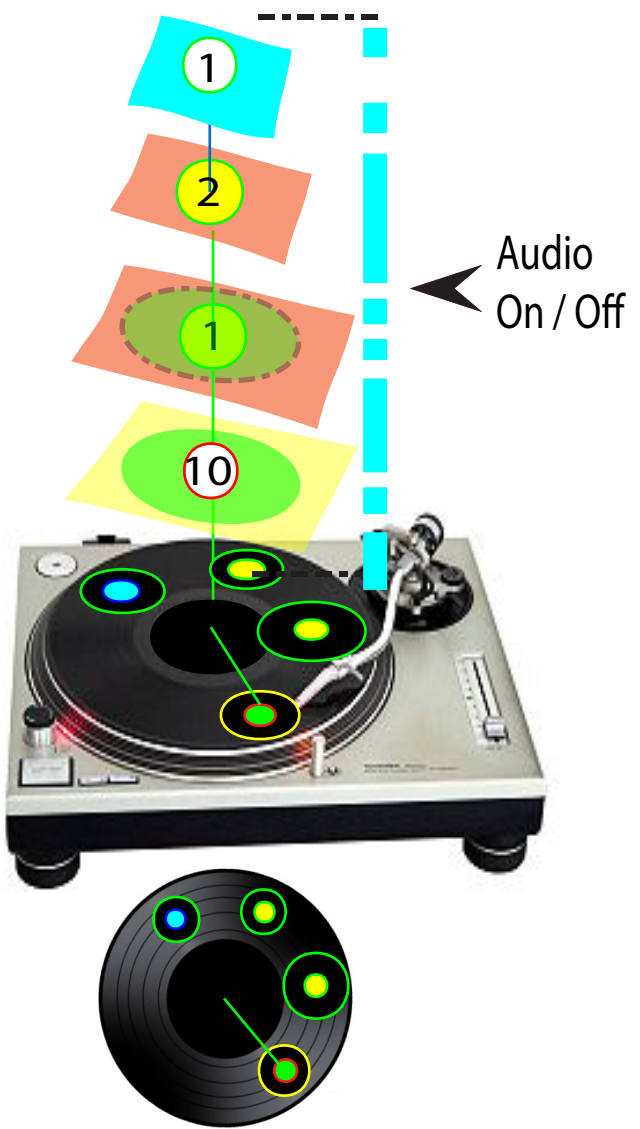
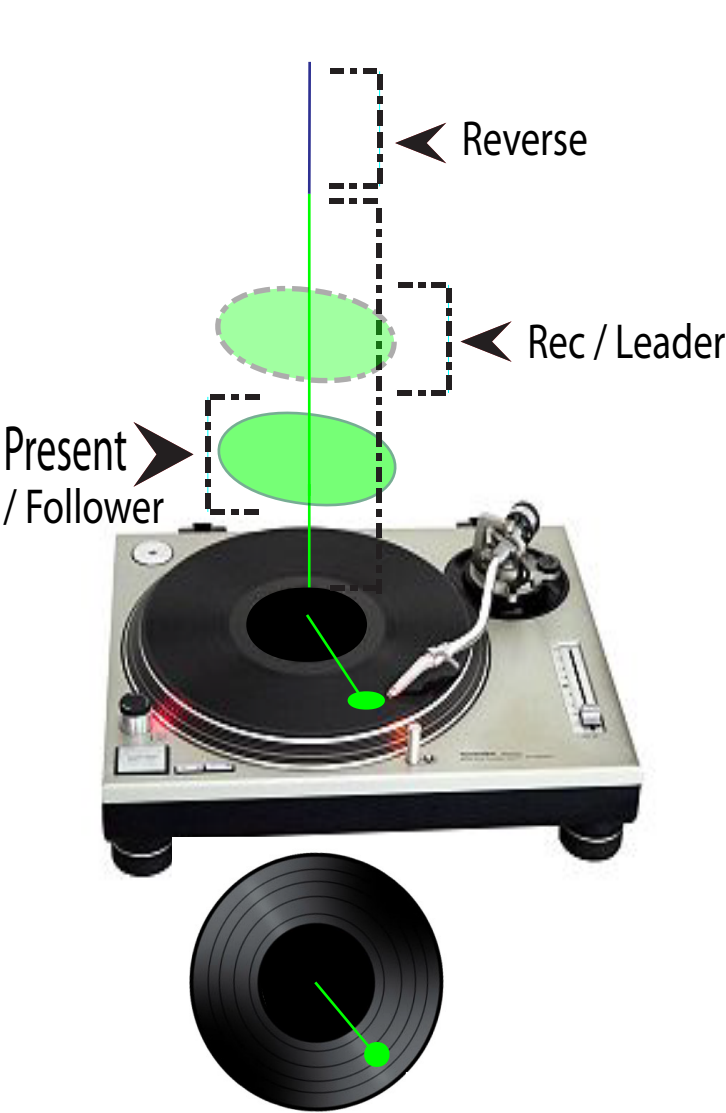


Freeze OR Reverse



Audio Crossfader

On / Off





JD NAME:

Play Test:

Do you play video games?

What is the genre of games you play? Digital or analog.

Have you ever played a rhythm game? Digital or analog.

"diamondsAndCode"

Play State | Rec State

Answer

Direction

Question

Omitter
Collection

Emitter
Distribte

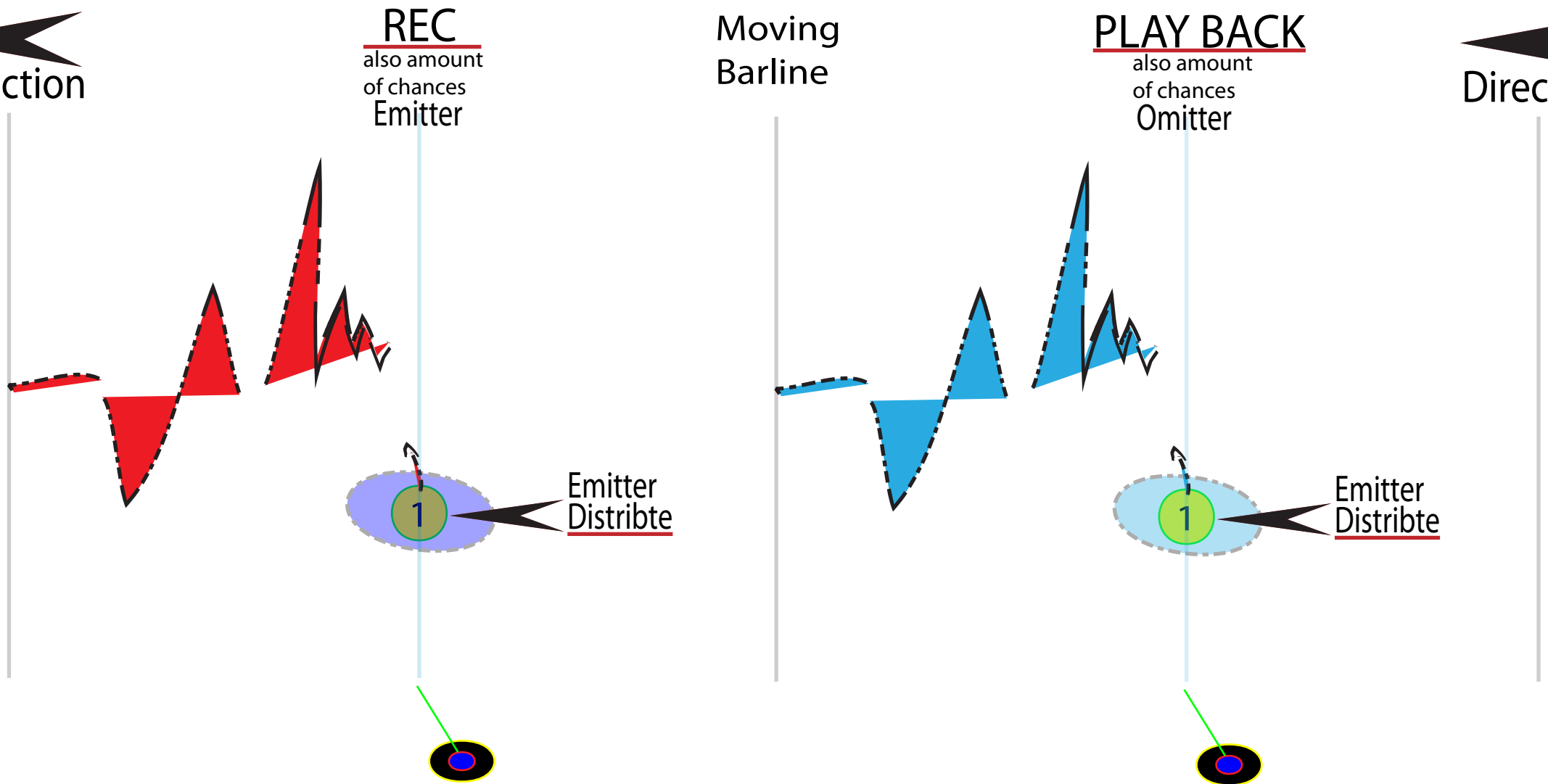


"diamondsAndCode"

Play State

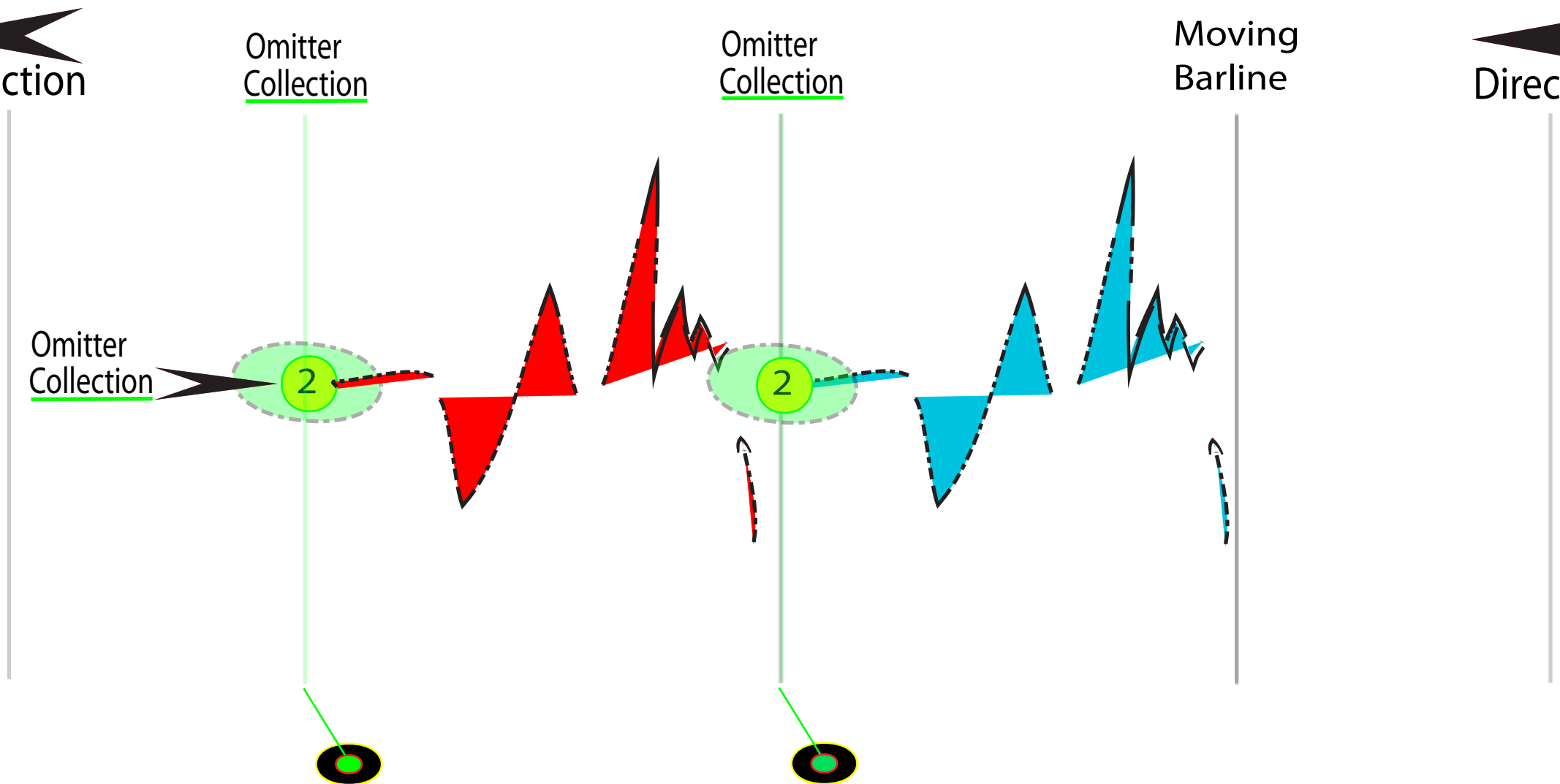
Rec State

Question



"diamondsAndCode"

Play State | Rec State
Answer



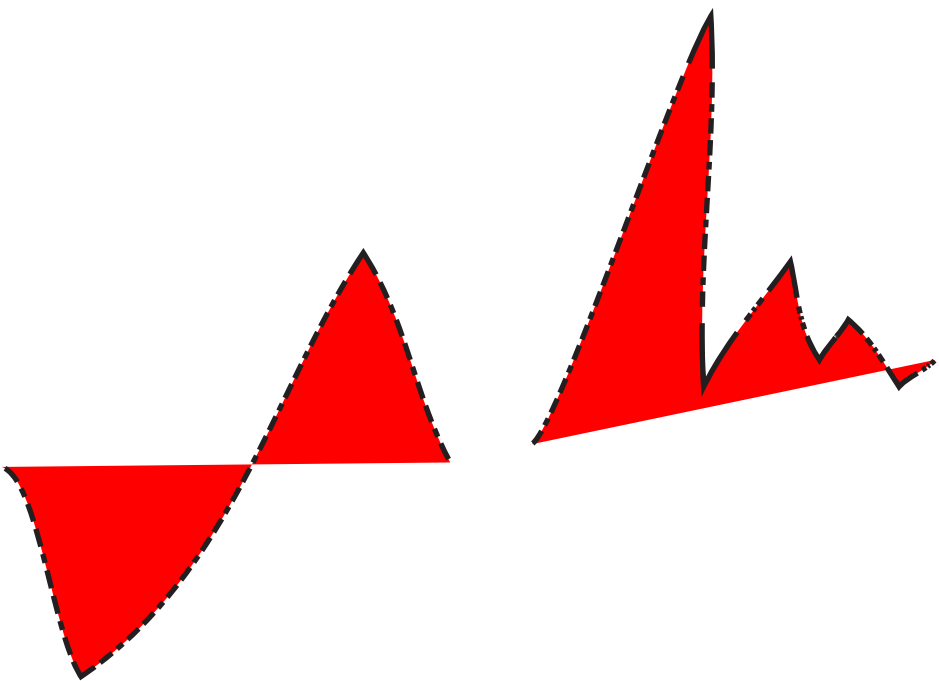
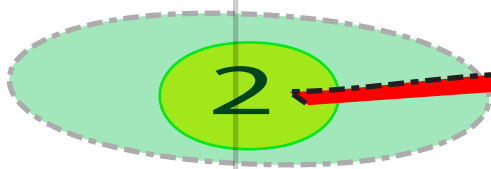
"diamondsAndCode"

Play State | Rec State

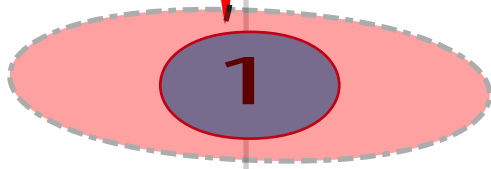
Wave



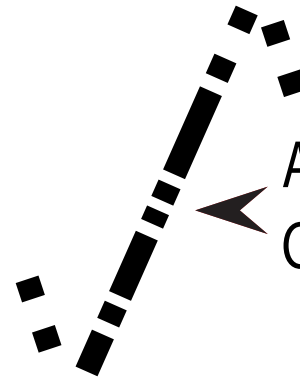
Omitter



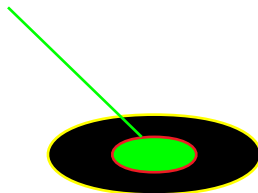
Emitter



Moving
Barline

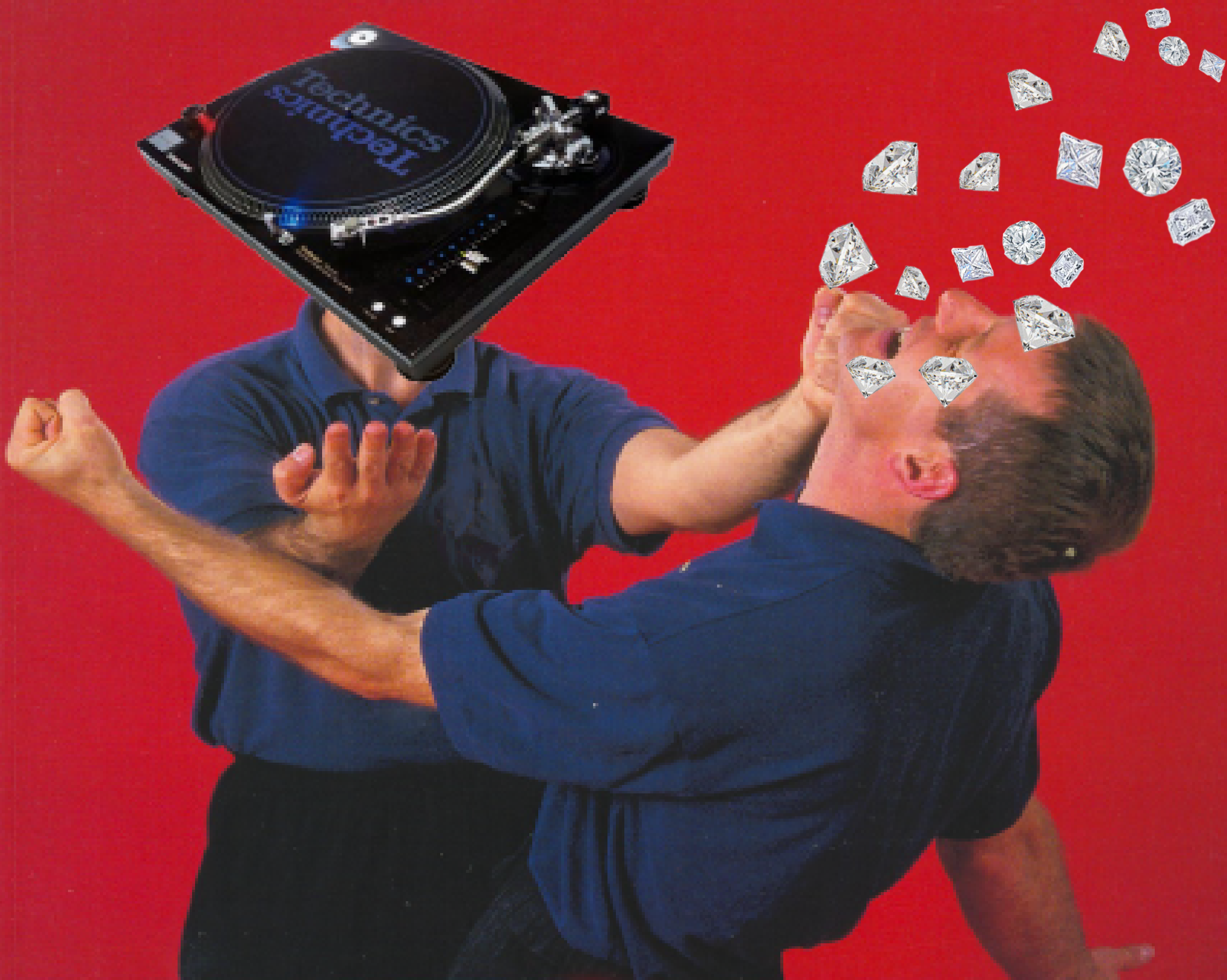


Audio
On / Off



SIMPLY...

diamondsAndCode



SIFU **TESUJI**

 Welcome to the digital Dojo 

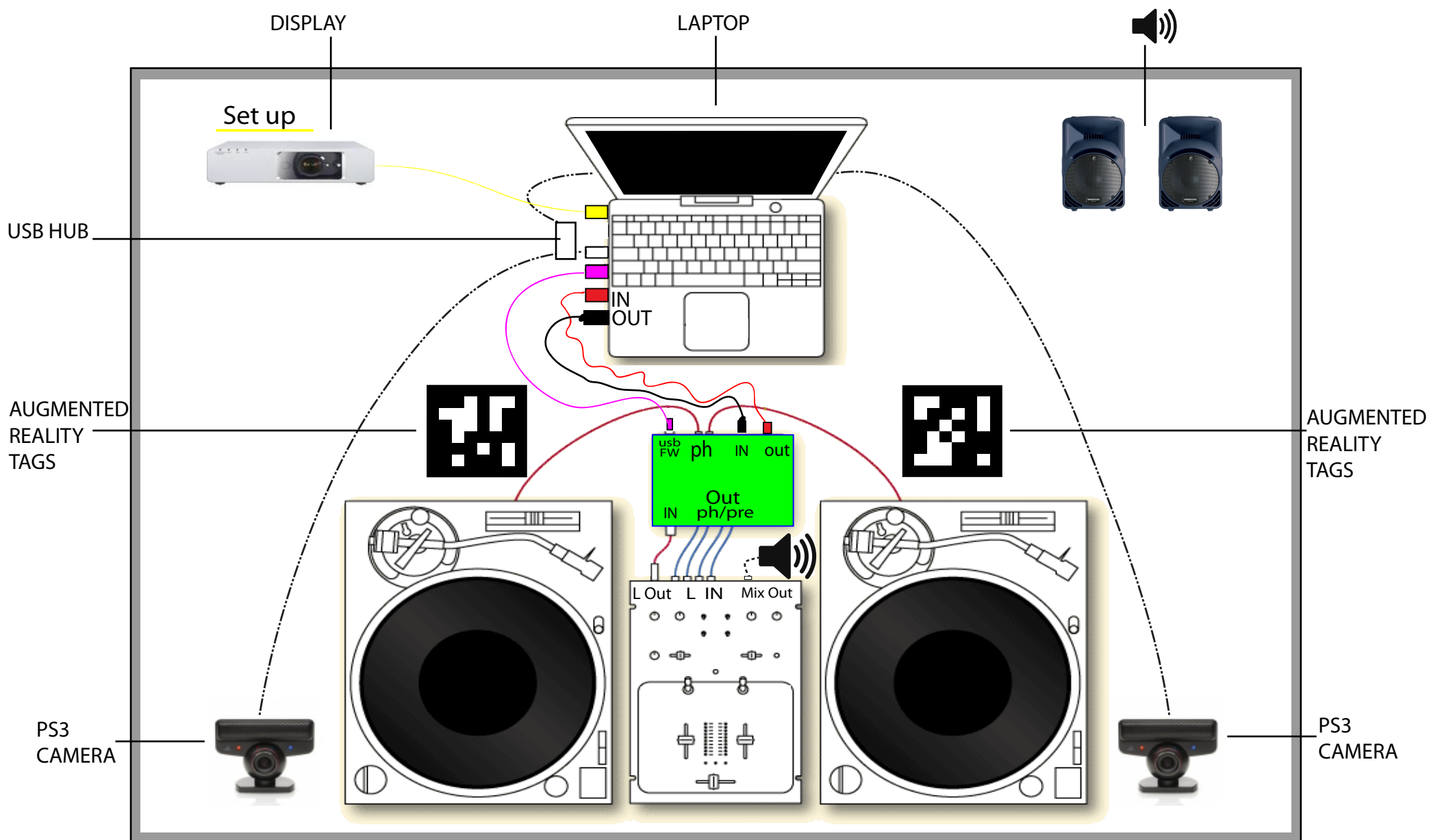


figure 6.0: Hardware Setup

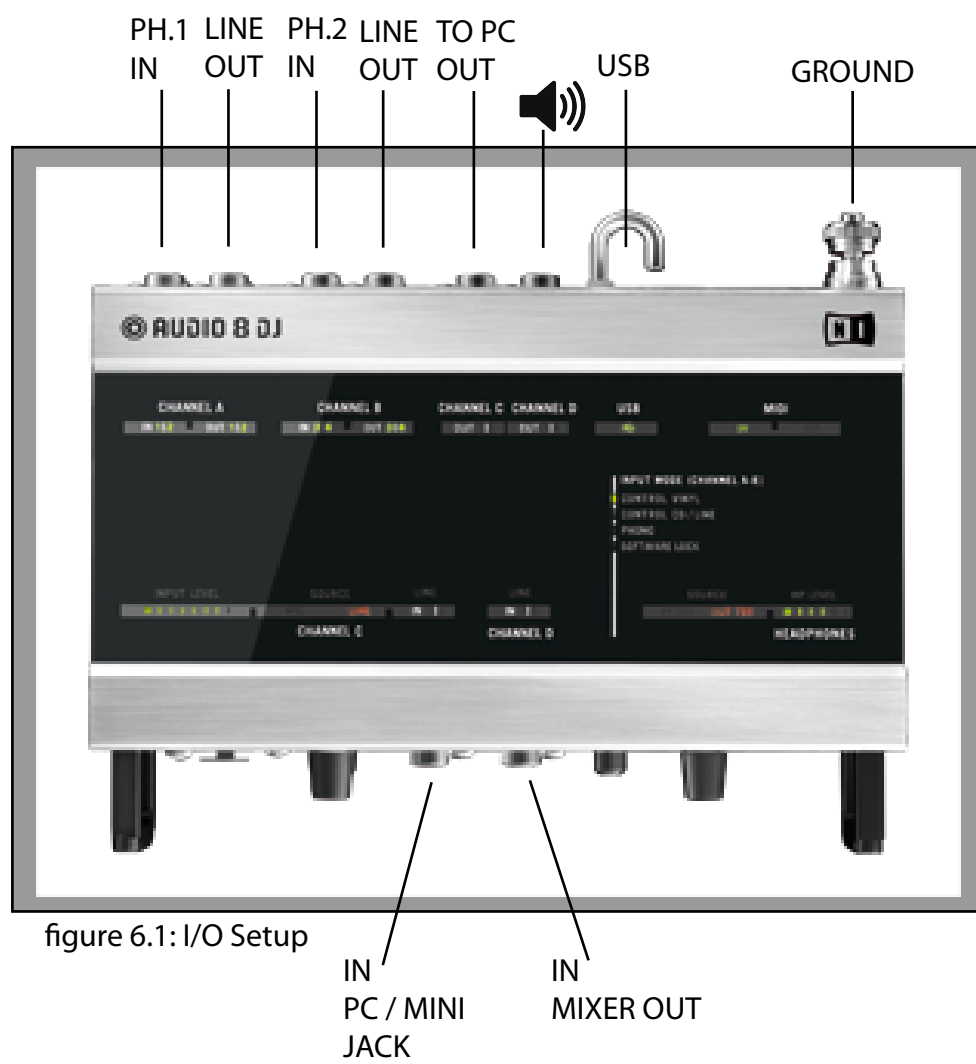


figure 6.1: I/O Setup



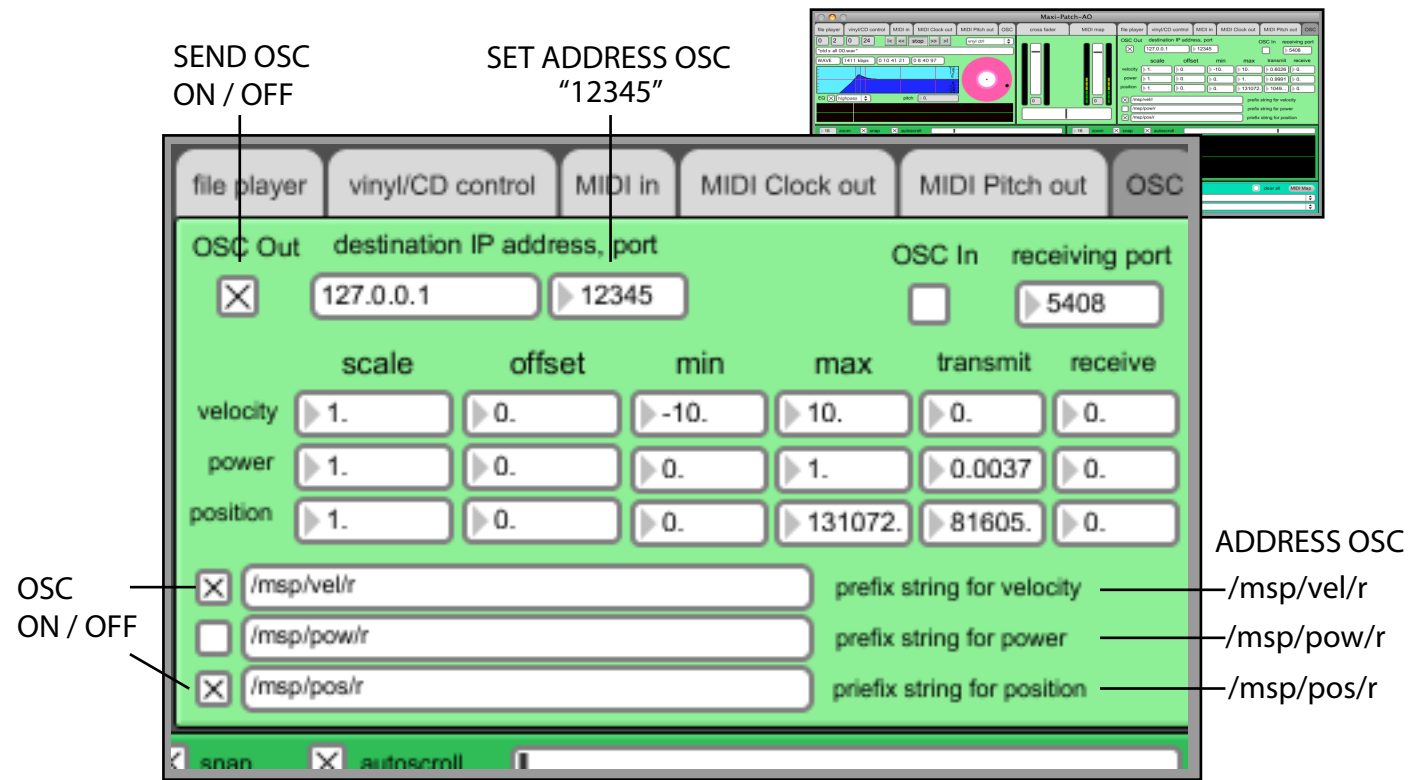


figure 7.0: OSC Maxi-Patch-AO

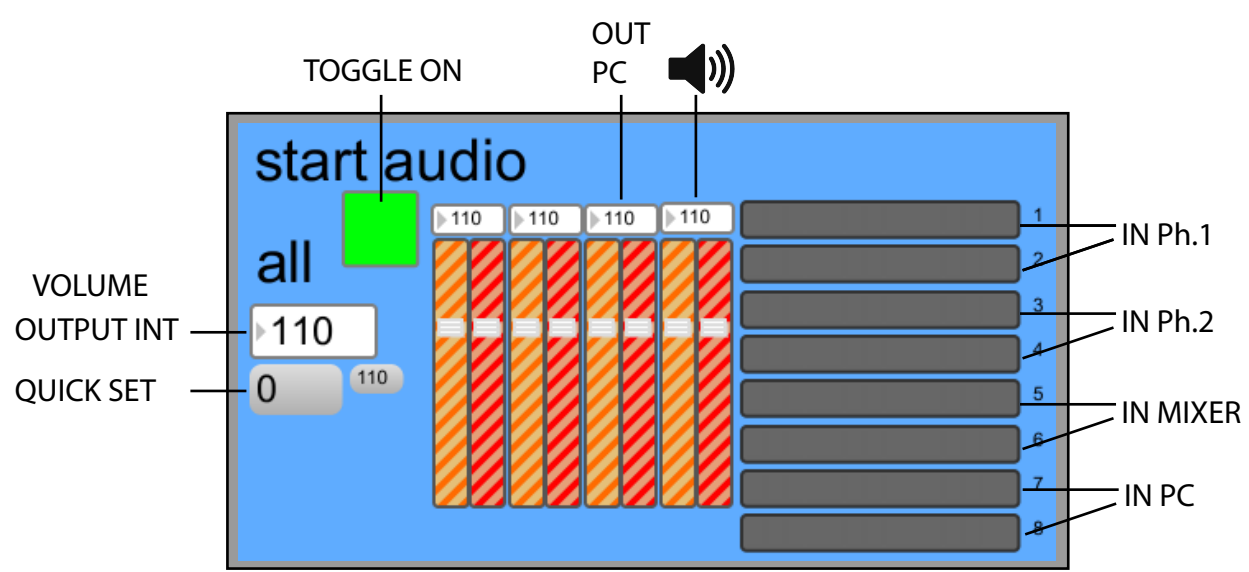


figure 7.1: Audio

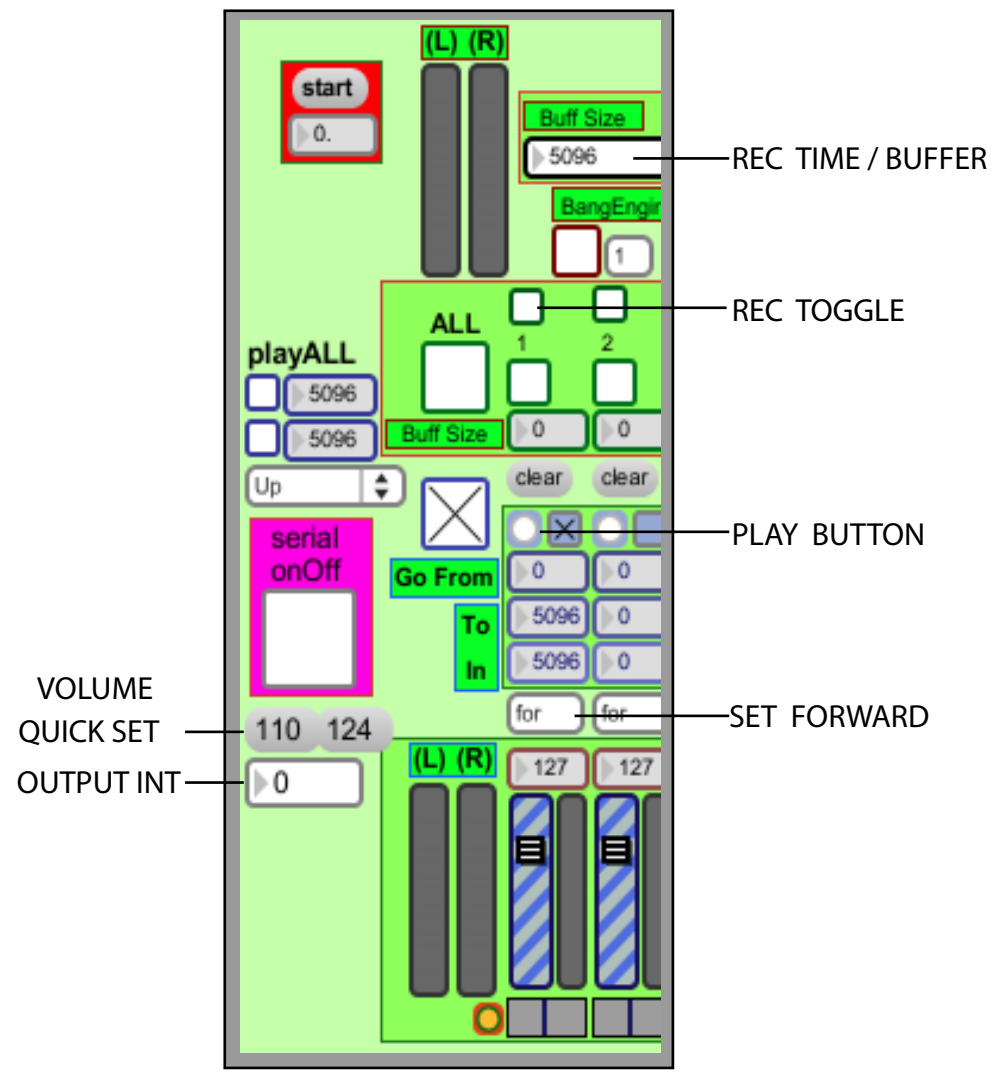


figure 7.2: Audio Rec / Play

THE GAME SCREEN



figure 1.0: Game Screen

TURNTABLE 1 / 2 — This is to let you exactly which turntable you are using.

COMPASS — This compass is stationary and telegraphs if you are in the correct zone.

EMITTER / OMITTER COURT POSITON 1 / 2 — This is your current position you push forward to move up or pull back to move down. (Warning!? This will only occur if the needles are on the records)

STATS MENU — This is where points are collected.

AUGMENTED REALITY — These are representations of the turntables maintained by augmented information. (without video display)

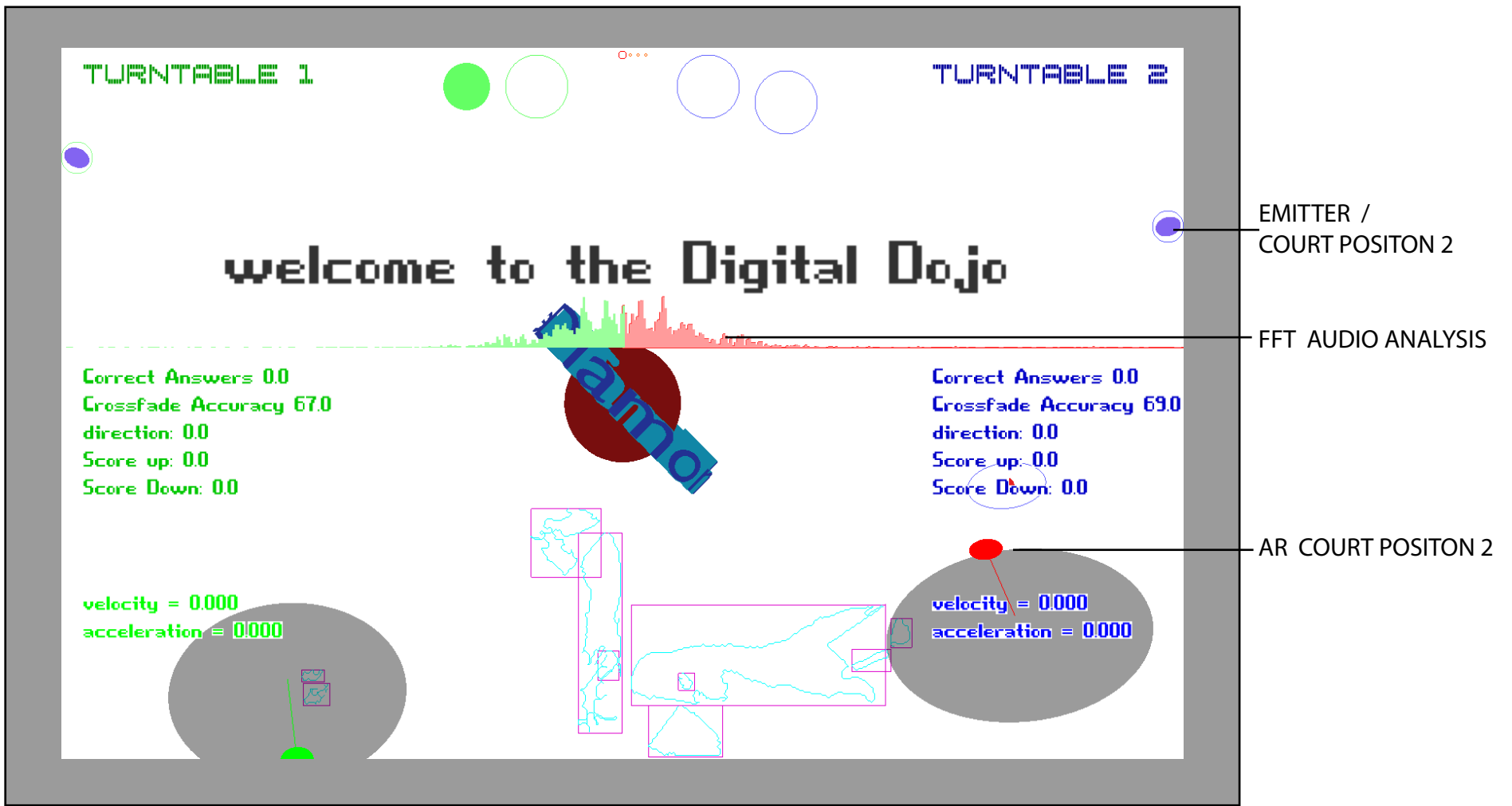


figure 2.0: Question Screen

FFT AUDIO ANALYSIS will turn red while
EMITTER / COURT POSITON is being recorded.

figure 2.1: Question FFT

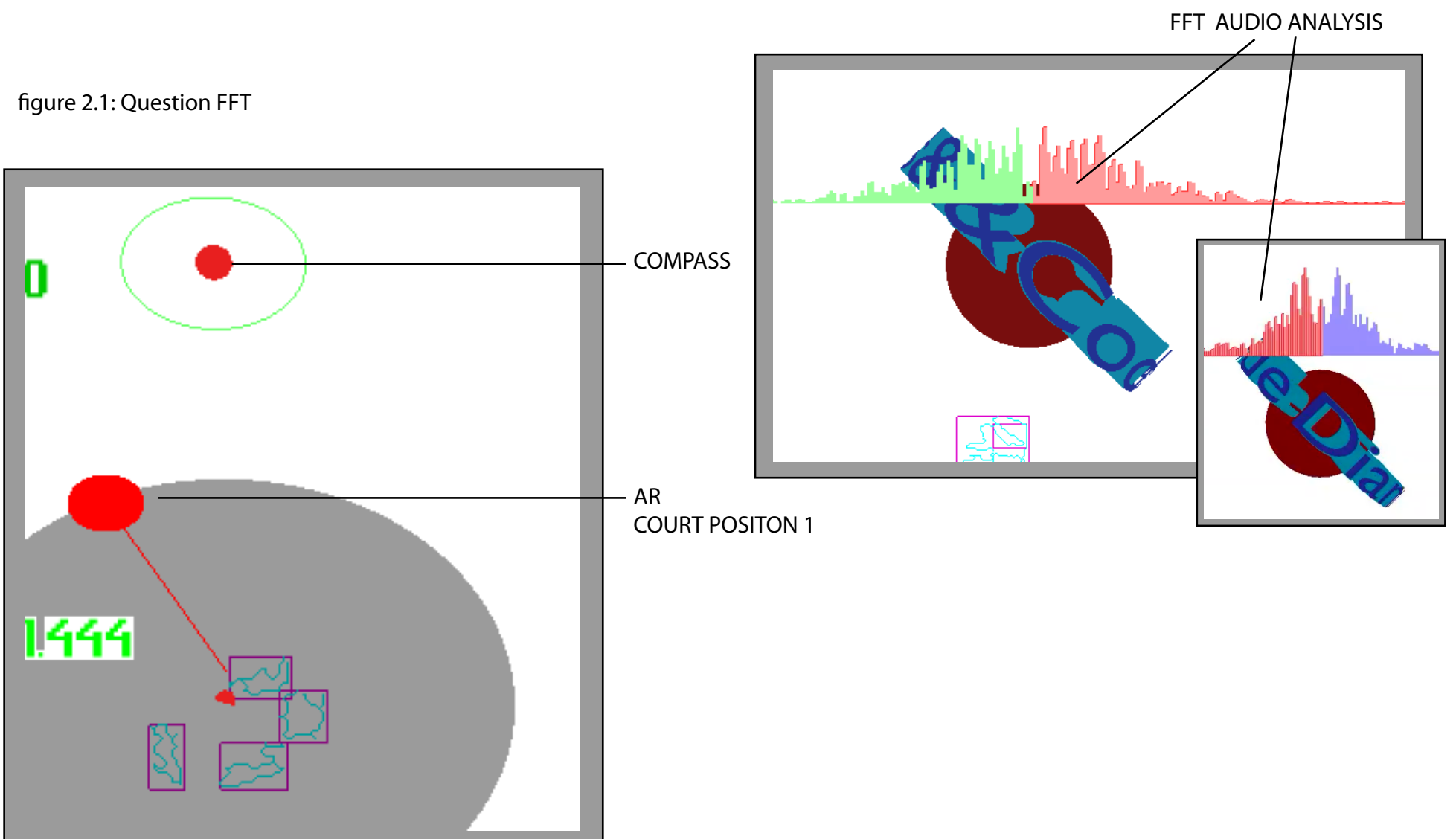


figure 2,2: Question AR

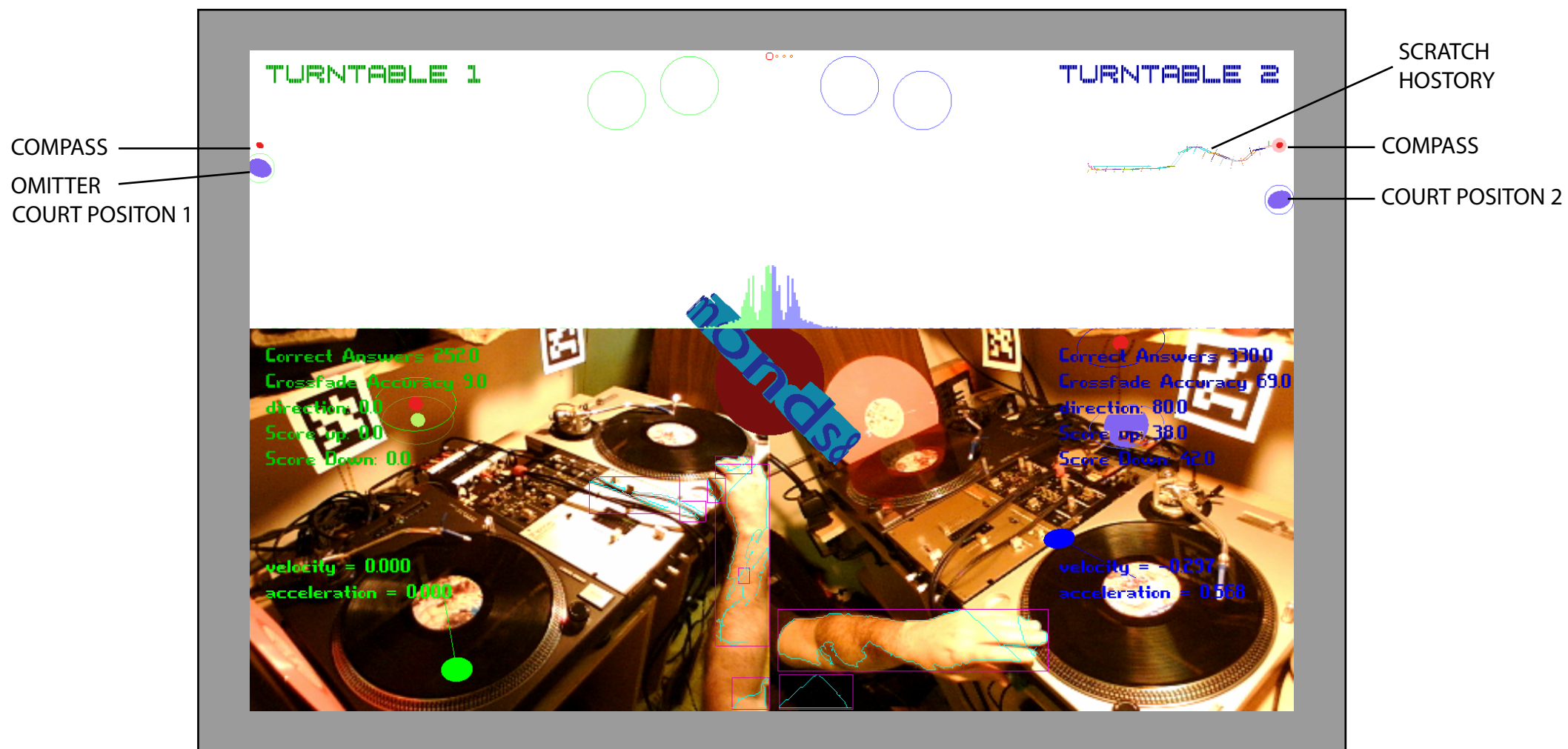


figure 3.0: Answer Screen

COMPASS ——— The COMPASS moves as the opposite turntable had while being recorded

OMITTER /
COURT POSITON 1 — You get points for having for the COURT POSITON on top of the COMPASS

SCRATCH
HOSTORY ——— The SCRATCH HOSTORY tells you where the COMPASS will be in the future

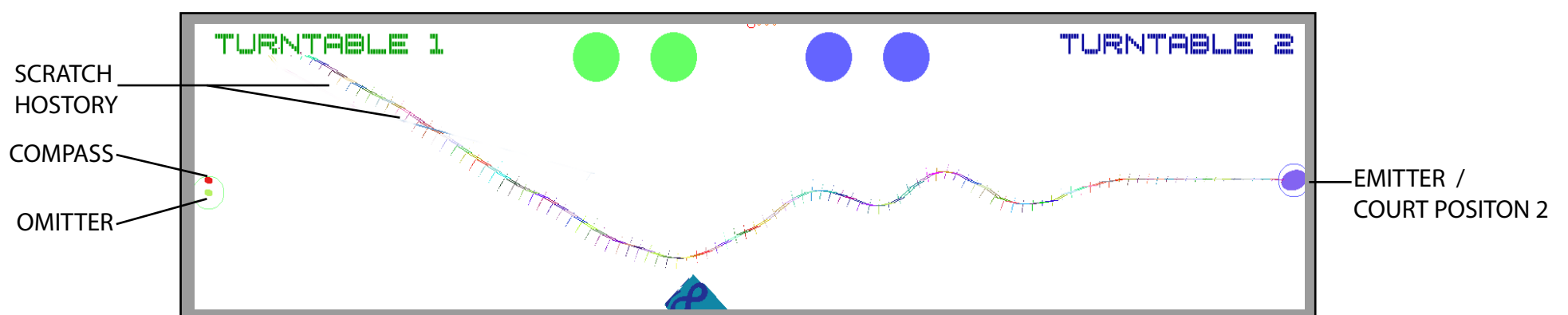


figure 3.1: Answer Top Screen

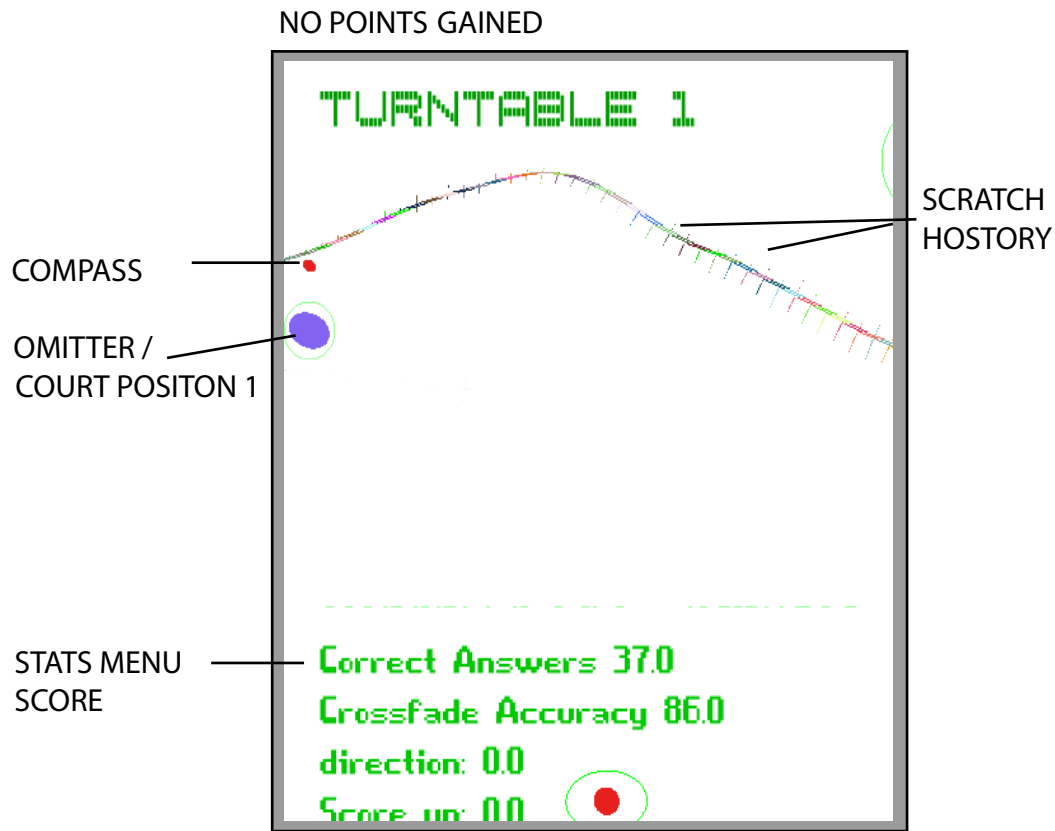


figure 4.0: Answer Omitter

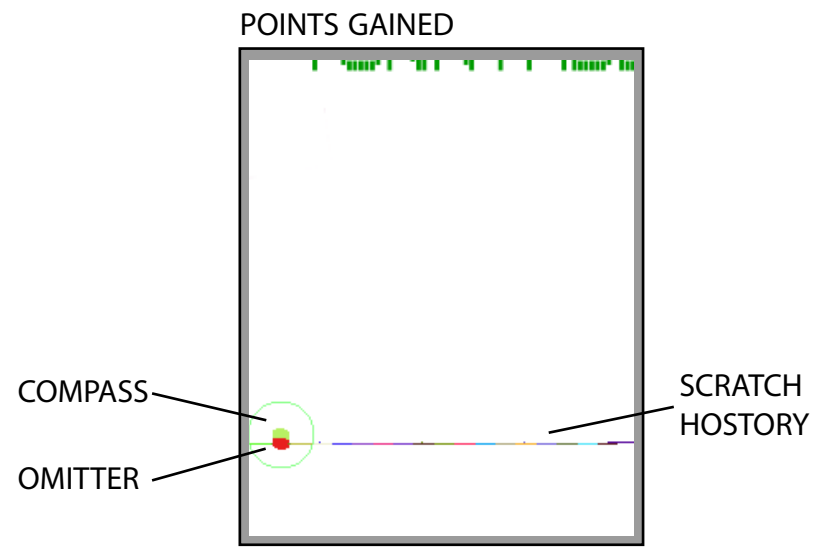


figure 4.1: Answer Omitter

COMPASS ————— The COMPASS moves as the opposite turntable had while being recorded

OMITTER / COURT POSITON 1 — You get points for having for the COURT POSITON on top of the COMPASS

SCRATCH HOSTORY ————— The SCRATCH HOSTORY tells you where the COMPASS will be in the future

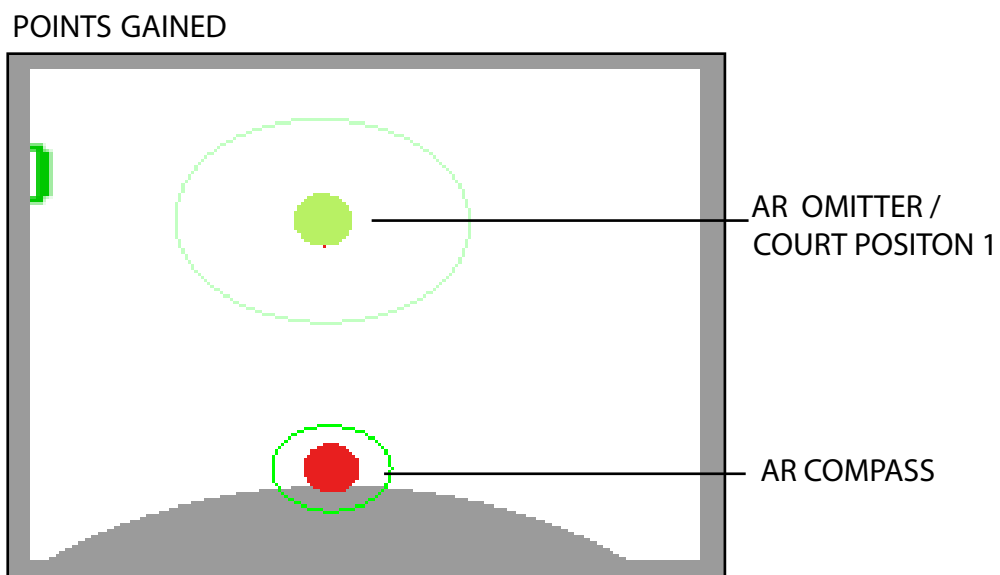


figure 4.2: AR Answer Omitter

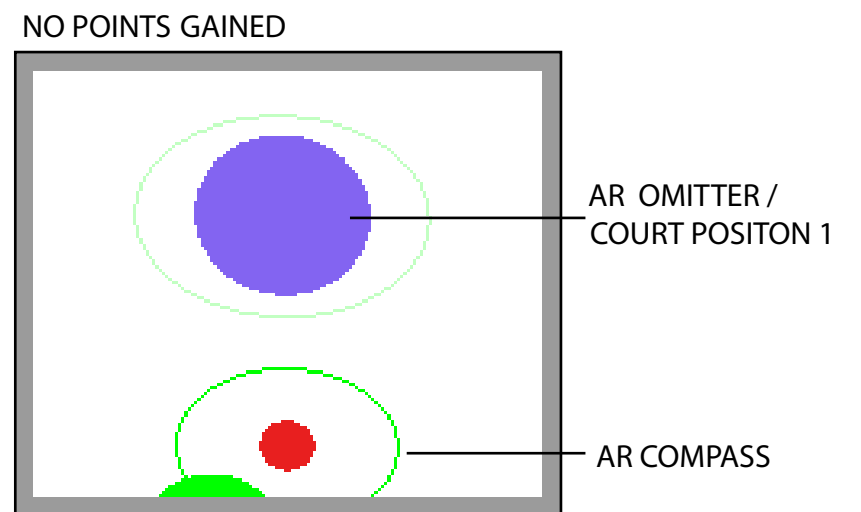


figure 4.3: AR Answer Omitter

If you choose to watch your own turntable during the Answer session of the game, you will find the augmented reality displays just as the two-dimensional heads up display does.

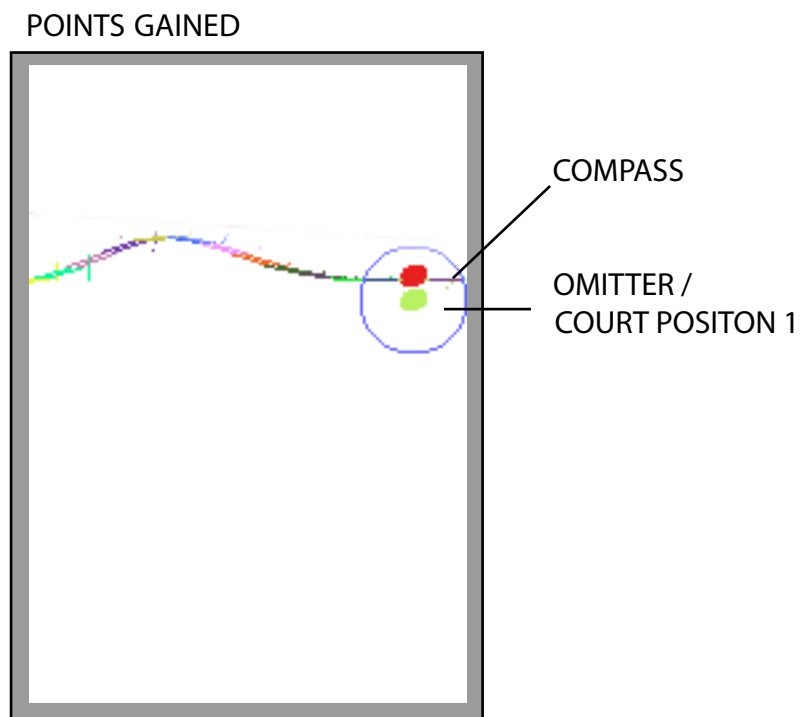


figure 5.0: Answer Score

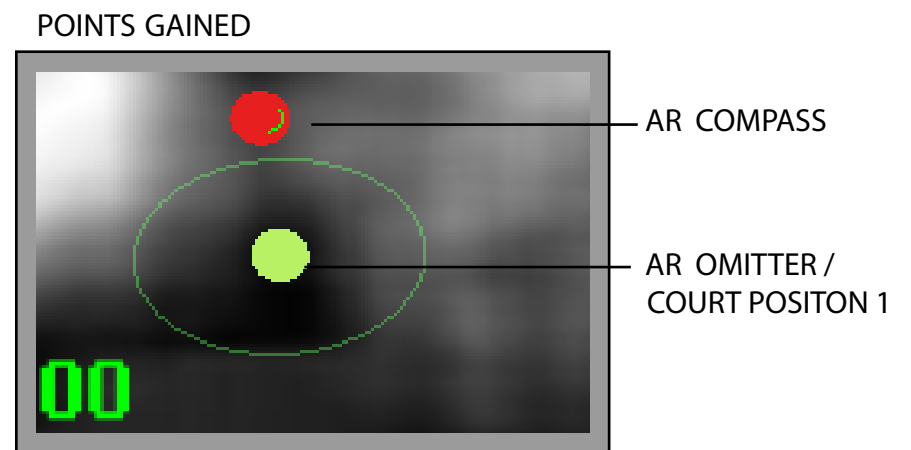


figure 5.2: AR Answer Omitter

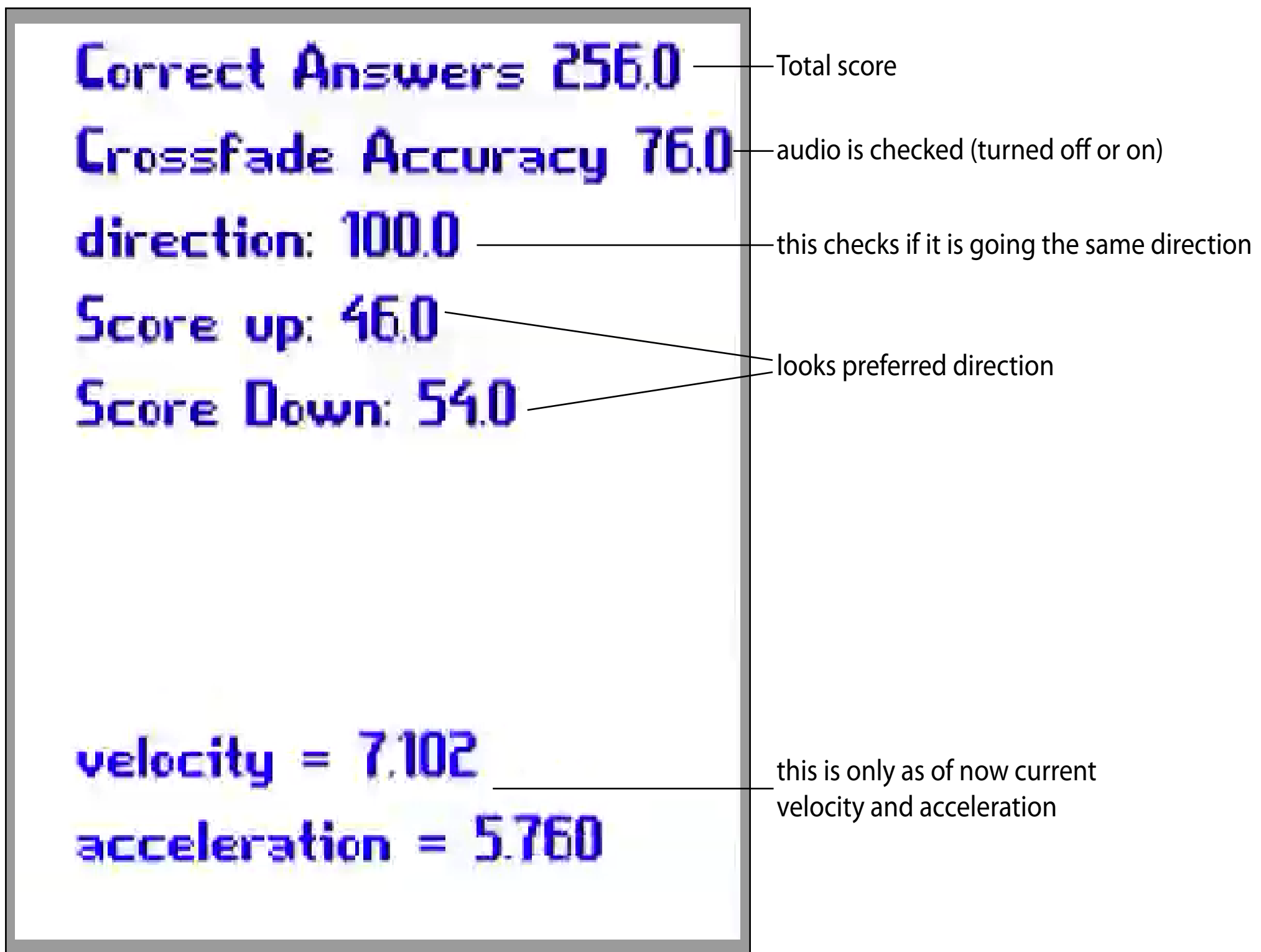


figure 5.1: Meun Score



JD NAME:

Play Test:

Do you play video games?

What is the genre of games you play? Digital or analog.

Have you ever played a rhythm game? Digital or analog.



JD NAME:

Play Test:

Does this experience with the game change what you think a DJ does?

Did the graphics in the game help you copy the other player scratch?

Did seeing the music graphically help?

What caused the music change?

How did you feel about the music?

Could you tell when it was your turn and had you ben give enough time to take your turn?



Could you tell on what criteria you were being scored?

Do you feel you have enough control on the game's outcome?

Did this improve your skills as a DJ?

Were there parts of the game you liked and what was it?

Were there parts of the game that you did not like, and what was it?

Do you play an instrument and if so what is it?
